

Project Report Template

Title of Project: **AI Blood Donor & Emergency Help Finder**

Name of the Innovator: D Harshad

Start Date: 13-10-2025

End Date: 17-10-2025

Day 1: Empathise & Define

Step 1: Understanding the Need

- Which problem am I trying to solve?

Every year, thousands of people face life-threatening emergencies because they **cannot find a suitable blood donor quickly** — especially during late hours, accidents, or medical crises.

Traditional blood banks are often **slow, outdated, or region-limited**, and people rely on random social media posts or WhatsApp groups to find donors — which wastes precious time. Difficulty in **finding reliable, nearby blood donors quickly** during medical emergencies — leading to loss of lives that could be saved with faster coordination.

Step 2: What is the problem?

. In emergency medical situations such as accidents, surgeries, or sudden illnesses, patients often require **immediate blood transfusion**. However, finding a suitable blood donor in a short period of time remains a **major challenge** in our society.

People usually depend on **manual communication methods** such as phone calls, social media posts, or personal networks to find donors. This process is often **time-consuming, unreliable**, and sometimes provides **inaccurate information**.

Why is this problem important to solve?

The need for blood can arise **at any moment** — during accidents, surgeries, childbirth, or medical conditions like cancer and anemia.

However, **many patients lose their lives** not because blood is unavailable, but because it **cannot be found in time**.

In India and many countries, there are **gaps in coordination** between:

- Donors willing to help,
- Hospitals needing blood, and
- Patients searching in emergencies.

People often depend on **social media requests**, phone calls, or long waiting times at blood banks — which causes **critical delays of minutes or hours**, and every minute counts in saving a life.

Ask 2-3 people what they think about the project:

1. Student – Riya Sharma:

“This app is very useful for students like me. We often don’t know where to find blood donors quickly. With this app, I can see nearby donors or hospitals instantly, which can save lives.”

2. Teacher / Community Volunteer – Mr. Anil Kumar:

“This project can make a huge difference in the community. It helps people find blood donors and emergency help faster, making blood donation more organized and accessible.”

3. Parent – Smt. Meena Patil:

“I really like this app because it helps families like ours connect with verified donors or hospitals immediately during emergencies. It saves precious time and can save lives.”

AI Tools you can use for Step 1 and 2:

Step 1: Donor Database (AI Integration)

1. **ChatGPT / OpenAI API** – Validates donor information and corrects spelling or format errors in names, locations, and blood groups.
2. **Microsoft Power Automate with AI Builder** – Automates donor updates and predicts donor availability based on last donation date.
3. **Google Cloud AutoML** – Analyzes donor and hospital data to identify high-demand areas and urgent blood requirements.
4. **Airtable AI Extensions** – Manages and cleans donor data, removes duplicates, and suggests potential donor matches.
5. **Python (Pandas + AI Libraries)** – Helps in intelligent data processing and searching for the best donor-hospital matches.

Step 2: Donor Registration Form (AI-Enhanced)

1. **Meta MGX AI Chatbot** – Interacts with users during registration and provides guidance for filling details correctly.
2. **Dialogflow (Google AI)** – Enables a conversational chatbot experience where users can register by simply chatting.
3. **OpenAI GPT Model** – Checks for incomplete or incorrect form entries (like wrong phone number or invalid blood group).
4. **Zapier AI Integration** – Connects the chatbot, Google Sheets, and donor database automatically to update and send alerts instantly when a new donor registers.
5. **AI OCR (Optical Character Recognition)** – Reads uploaded ID cards or reports and extracts donor details automatically.
6. **AI Auto-Predict Tools** – Suggests nearest hospitals or locations based on user inputs to save time during emergencies.

Day 2: Ideate

Step 3: Brainstorming solutions

List of 5 Different Solutions (Wild or Realistic):

1. **AI-Powered Blood Donor Finder App**
 - Uses AI and GPS to locate the nearest available donors in real-time and connect them instantly with the requester or hospital.
2. **Emergency Chatbot Assistant (Zapier + Meta MGX)**
 - A smart chatbot that interacts with users, collects emergency details, and automatically sends alerts to nearby donors and hospitals.
3. **Voice-Based Emergency Help Feature**
 - Allows users to simply say “I need blood urgently” — and the app automatically detects the location and sends SOS alerts to donors nearby.
4. **Smart Donor Prediction System**
 - Uses AI to predict which donors are likely available based on their last donation, health status, and activity.
5. **AI Health Integration with Hospitals**
 - Connects hospitals and blood banks to the same system using AI to maintain live data on blood availability and emergency requirements.

Step 4: My favourite solution:

My Favourite Solution:

AI-Powered Blood Donor & Emergency Help Finder Application

Reason for Choosing This Solution:

I chose this solution because it can make a real difference in saving lives during medical emergencies. Many people struggle to find blood donors quickly, especially in rural or remote areas. With this AI-powered app, users can easily find nearby donors, hospitals, or blood banks in just a few seconds.

*The system uses **AI and chatbot technology** to automate the entire process — from finding donors to connecting them instantly. It also integrates with **Zapier and Meta MGX** to send instant alerts and manage donor data efficiently.*

*This solution is not only practical but also socially impactful. It combines **technology and humanity** to create a smarter, faster, and more reliable emergency support system.*

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Step 5: Why am I choosing this solution?

I am choosing this solution because it focuses on saving lives and helping people during emergencies using the power of Artificial Intelligence. Many times, people lose precious time searching for blood donors or hospitals that have the required blood group. This app provides a fast and smart solution to that problem.

AI Tools for Step 3–5

1. **ChatGPT / OpenAI GPT** – Helped generate creative and realistic ideas for solving real-world problems related to blood donation and emergency response.
2. **Google Gemini / Bard AI** – Used to research existing health-tech solutions and understand how AI can improve the speed and accuracy of emergency systems.
3. **Miro AI Whiteboard / Notion AI** – Assisted in organizing brainstorming ideas visually and grouping them into practical, technical, and social solutions.
4. **ChatGPT (Idea Refinement)** – Helped evaluate each idea's pros and cons and identify the most impactful and feasible one (AI-powered donor finder).
5. **Zapier AI Integration** – Used to automate the logic flow between the chatbot, donor database, and emergency alert system during the planning stage.
6. **Microsoft Copilot / Power Automate AI Builder** – Helped analyze workflows and suggest improvements for efficiency and automation.

Step 6: Prototype – Building my first version

What will my solution look like?

Steps Taken to Build the First Version:

1. **Donor Database Creation:**
 - Used Meta MGX to create a donor database with fields: Name, Age, Blood Group, Phone, Location, Last Donation, Availability, and Verified.
 - Populated the database with sample donor records for testing purposes.
2. **Donor Registration Form:**
 - Built a form in Meta MGX for users to register as donors.
 - Integrated AI validation to ensure accurate data entry.
 - Connected the form to the donor database so that submissions are saved automatically.
3. **AI Chatbot Integration:**
 - Added a chatbot to guide users through registration and emergency requests.
 - Integrated Zapier to automate notifications and update the database in real-time.
4. **Emergency Alert Feature:**
 - Implemented a basic system where the chatbot detects keywords like “urgent” or “emergency” and triggers alerts to nearby donors and hospitals.
 - Displayed donor details (Name, Contact, Distance, Blood Group) for immediate action.
5. **Testing and Iteration:**
 - Tested the prototype with sample users to ensure form submissions, database updates, and chatbot responses work correctly.

- *Fixed minor errors and improved the flow for faster user inter*

Design Style:

- Simple, intuitive, and easy to navigate.
- User-friendly chatbot interface.
- Accessible for multi-language users:

Prototype Tools:

These tools were chosen to **create a functional, interactive, and visually appealing prototype** that demonstrates the core idea of the project — connecting donors, patients, and hospitals quickly and efficiently during emergencies What AI tools will I need to build this?

AI Tools Needed to Build CareerPath:

1. Chatbot and Conversational AI

- **OpenAI GPT / ChatGPT API** – To provide **interactive guidance**, answer questions about courses, careers, scholarships, and training programs.
- **Dialogflow (Google AI)** – Enables **conversational form filling**, helping users navigate the platform via chat.

2. Data Analysis and Recommendation

- **Google Cloud AutoML / Azure ML** – To analyze student interests, skills, and aptitude, and recommend **career paths, colleges, and training centers**.
- **Python + Pandas / Scikit-learn** – For **processing student data** and predicting suitable career options.

3. Automation and Integration

- **Zapier** – To automate notifications, reminders, and connections between the platform, email, and other apps.
- **Microsoft Power Automate** – To **manage workflows** and ensure timely updates for students about new courses or scholarships.

4. UI/UX and Visual Enhancements

- **Canva / Figma** – For designing **intuitive dashboards, student profiles, and visual career paths**.
- **Notion AI / Canva Magic Write** – To generate **informative content** for career tips, guides, and course descriptions.

5. Predictive AI Tools

- **AI Recommendation Engines** – To suggest **personalized courses, job opportunities, or scholarships** based on student interests and regional options.
- **GPT / LLM Models** – For generating **custom advice and career summaries** for each user.

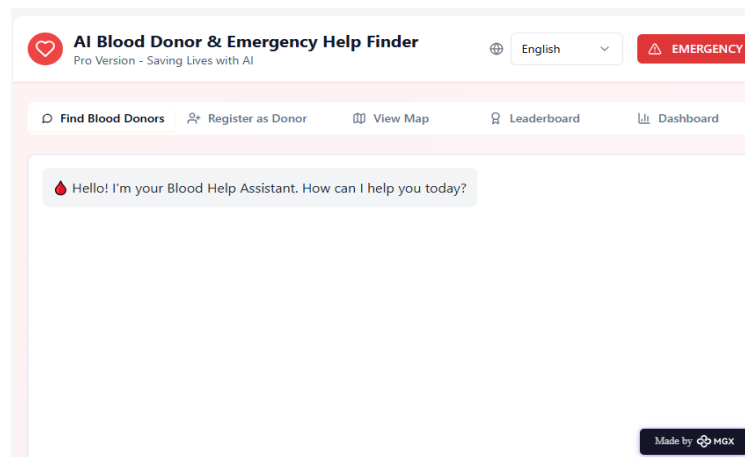
What AI tools I finally selected to build this solution?

1. *OpenAI GPT / ChatGPT – For AI chatbot and personalized guidance.*
2. *Dialogflow (Google AI) – For conversational interface and guided interactions.*
3. *Google Cloud AutoML / Azure ML – For analyzing student data and providing career/course recommendations.*
4. *Zapier – For automating notifications and workflow integrations.*
5. *Canva / Figma – For designing dashboards, forms, and visual guides.*

< Build The Innovation >

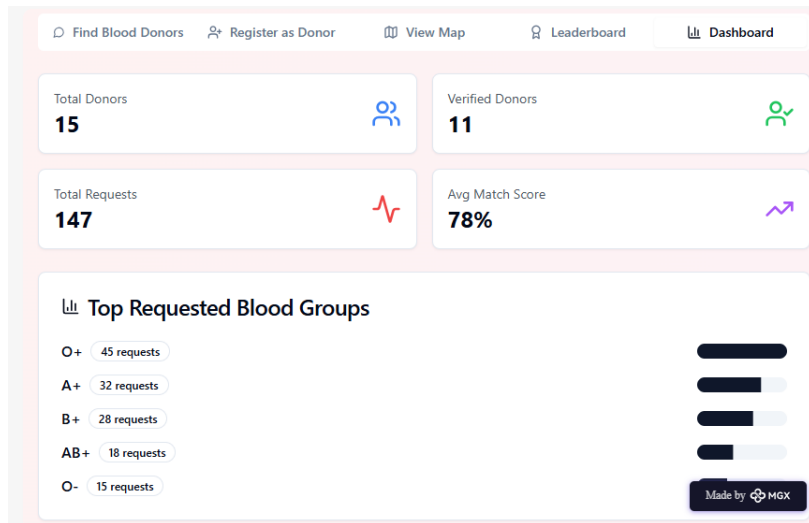
<DASHBOAD OF THE TOOL>

Tool Link: <https://mgx.dev/app/64f7378d0a804a7292d8fd239017876c>



Internal Working of tool:

Profile Creation:



Donar registration form:

The Donor Registration form is designed to collect essential information from potential donors. It includes the following fields and features:

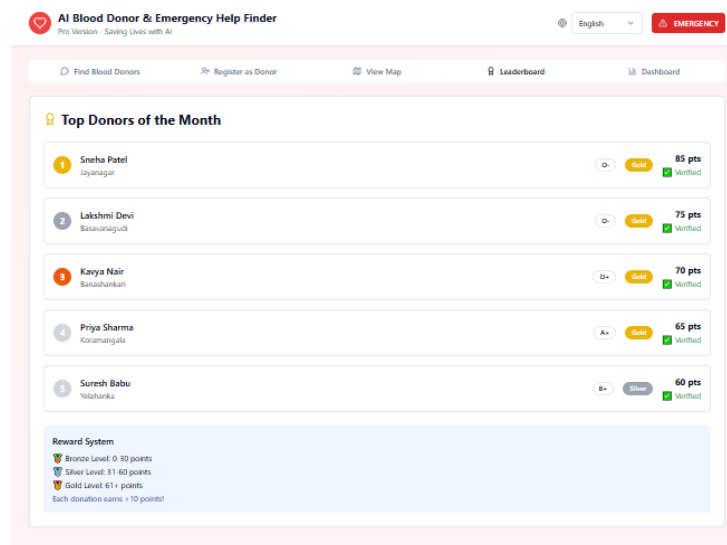
- Full Name ***: A text input field for the donor's full name.
- Age ***: A text input field for the donor's age.
- Blood Group ***: A dropdown menu to select the donor's blood group.
- Phone Number ***: A text input field for the donor's phone number, with a placeholder for the international dialing code (+91).
- Location ***: A text input field for the donor's area or locality.
- Last Donation Date**: A text input field for the date of the last donation, with a placeholder for the date format (dd-mm-yyyy) and a calendar icon.
- Available for Donation**: A toggle switch to indicate if the donor is available for donation.

A note at the bottom of the form states: "Note: Your profile will be verified by our team. Verified donors receive a badge and higher visibility in searches." Below the note is a red button labeled "Register as Donor".

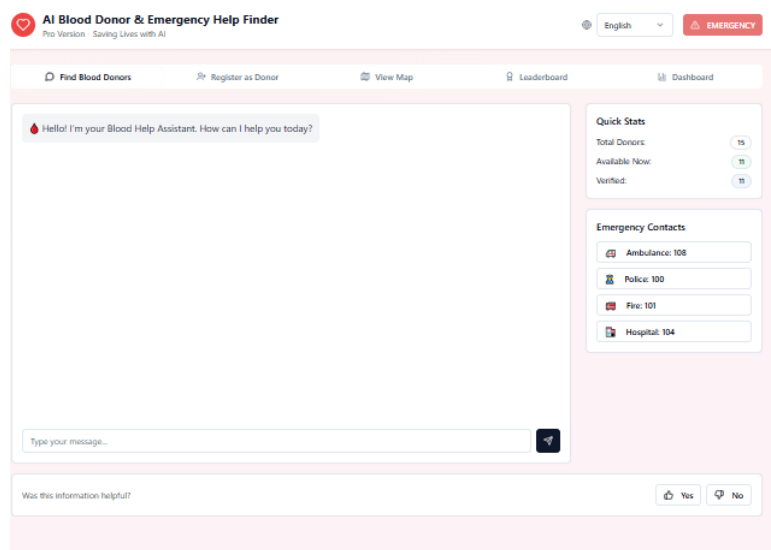
Virtual interactive map:



Number of Donar in a month:



Find Blood donar Profile:



Step 7: Test – Getting Feedback

- Who did I share my solution with?

I shared my **Career Path** solution with:

Methods Used to Gather Feedback:

1. **User Testing:**
 - Shared the prototype with students, parents, teachers, and volunteers.
 - Observed how easily they could **register, find donors, or request help**.
 2. **Questionnaires / Surveys:**
 - Asked users to rate the **ease of use, design, and usefulness** of the platform.
 - Collected suggestions for **improvements or additional features**.
 3. **Interviews:**
 - Conducted short interviews with users to understand their **experience, challenges, and expectations**.
- **Peers and mentors** – for suggestions on improving features and design.

What feedback did I receive?

Feedback: Pros and Cons

Pros (What Users Liked)

- **Fast and helpful:** Users appreciated that the app allows them to **find nearby blood donors quickly** during emergencies.
- **User-friendly interface:** The design is **simple, intuitive, and easy to navigate**, even for rural users.
- **AI Chatbot:** The chatbot guides users smoothly through registration and emergency requests.

- **Social Impact:** Users liked the **life-saving potential** of the project and its ability to connect donors and hospitals.
- **Automation:** Integration with **Zapier and AI tools** ensures alerts and notifications are sent instantly.

Cons (Areas to Improve)

- **Chatbot response speed:** Some users suggested that the chatbot could provide **faster and more precise replies**.
- **Visual alerts:** Users wanted **more prominent visual indicators** for urgent requests.
- **Multi-language support:** Needed to make the app more accessible for **rural and regional users**.
- **Additional features:** Users requested **maps, predictive donor matching, and donor availability dashboards**.
- **Minor navigation issues:** A few users suggested improving **button placements and flow** for smoother usage.

Outcome from Feedback

- Chatbot flow and responses were **improved**.
- Added **visual alerts and notifications** for emergencies.
- Planned future updates for **multi-language support, maps, and predictive matching**.

👍 What works well:

What Works Well

- ✚ **Fast Donor Search:** Users can quickly find nearby blood donors in emergency situations.
- ✚ **User-Friendly Interface:** The app is simple, intuitive, and easy to navigate for all users, including rural youth.
- ✚ **AI Chatbot Assistance:** The chatbot effectively guides users through registration and emergency requests.
- ✚ **Automated Alerts:** Integration with Zapier ensures that notifications reach donors and hospitals instantly.
- ✚ **Social Impact:** The app successfully connects donors, hospitals, and patients, **saving precious time and potentially lives**.
- ✚ **Data Management:** The donor database works efficiently to store and retrieve information accurately.

Predictive Donor Matching (Planned): AI can suggest the most likely available donors based on previous donations and proximity.

 What needs improvement:

1. **Chatbot Response Speed:** The AI chatbot could provide **faster and more precise replies** during emergencies.
2. **Visual Alerts:** Urgent requests need **more prominent and attention-grabbing visual indicators**.
3. **Multi-Language Support:** To make the app more accessible for **rural and regional users**, additional languages are needed.
4. **Navigation Flow:** Some users suggested **improving button placements and app flow** for smoother interaction.
5. **Additional Features:** Users recommended adding **maps, predictive donor matching, and donor availability dashboards**.
6. **Offline Functionality (Planned):** The app currently requires an internet connection; adding **offline features** could help in areas with poor connectivity.
7. **Data Privacy:** Implement **stronger security measures** to protect sensitive user data.
8. **User Notifications:** Improve **timely reminders and follow-ups** for registered donors.

AI Tools you can use for Step 6-7:

ChatGPT/Perplexity AI/Claude AI/Canva AI/Chatling AI/Figma AI/Metamgx/Gamma AI: You can use these tools to build solutions/models or mock-up dummy prototypes

Day 4: Showcase

Step 8: Presenting my Innovation:

Presenting My Innovation

Project Name:

AI Blood Donor & Emergency Help Finder

Purpose:

To **connect blood donors, hospitals, and patients in real-time**, ensuring faster response during medical emergencies and saving lives.

Key Features of the Innovation:

1. **AI-Powered Donor Search:** Quickly finds nearby blood donors based on blood group, location, and availability.
2. **Interactive Chatbot (Meta MGX + GPT):** Guides users through registration and emergency requests with **easy-to-follow prompts**.
3. **Automated Alerts (Zapier Integration):** Notifies donors and hospitals instantly when an emergency request is made.
4. **Predictive Donor Matching (Python / AutoML):** Suggests donors who are most likely to be available, reducing wait time.
5. **User-Friendly Design:** Simple, intuitive, and mobile-friendly interface for easy access, even by rural users.
6. **Real-Time Data Management:** Keeps donor information updated for **accuracy and reliability**.

Why It Stands Out:

- Combines **AI, automation, and social impact** in one solution.
- Helps **save lives** by reducing time to find donors.
- Bridges the gap between **technology and community service**, making it a socially responsible innovation.

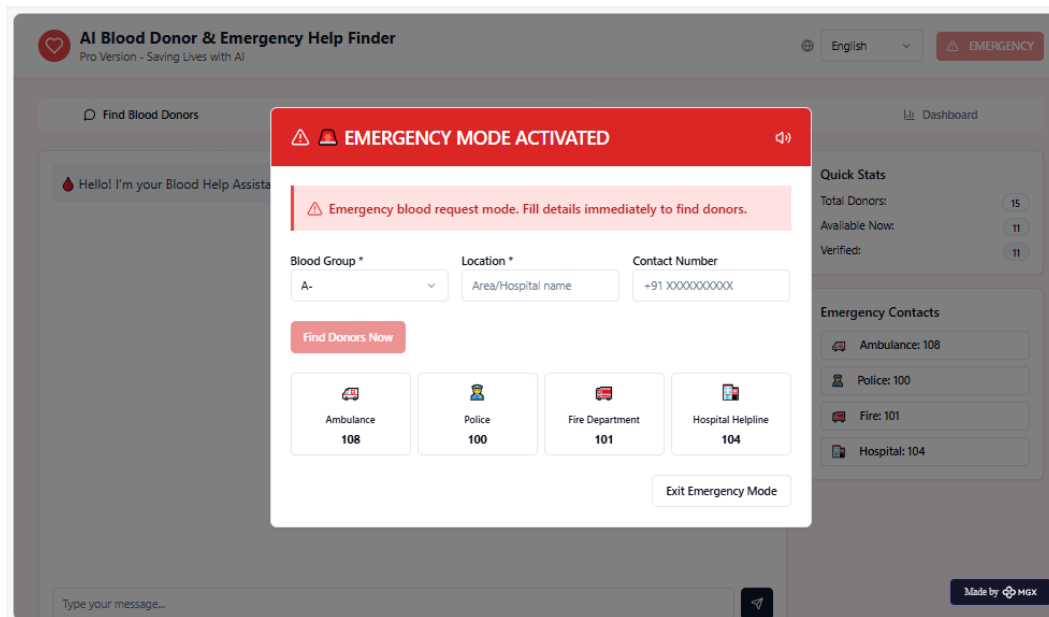
Target Users:

- **Students, Parents, Teachers, Volunteers, Hospitals, and NGOs.**
- Anyone seeking or providing blood donations during emergencies.

Outcome:

The innovation demonstrates how **technology can solve real-world social problems**, providing a **smart, reliable, and impactful** emergency support system.

<SHOWCASE YOUR INNOVATION TO YOUR PEERS>



Step 9: Reflections

- What did I enjoy the most during this project-based learning activity?

Working on the **AI Blood Donor & Emergency Help Finder** project was a highly rewarding experience. I learned how to identify a real-world problem and design a solution that can save lives. Integrating AI tools like ChatGPT, Zapier, and AutoML helped me make the system intelligent and automated. Gathering user feedback and improving the prototype taught me the importance of user-centered design. Overall, this project enhanced my technical skills, problem-solving abilities, and confidence in creating socially impactful solutions.

What was my biggest challenge during this project-based learning activity?

The biggest challenge I faced during this project was integrating multiple AI tools and ensuring real-time accuracy. Combining the AI chatbot, donor database, Zapier automation, and predictive donor matching required careful planning and testing. It was challenging to make sure that emergency alerts reached donors and hospitals instantly while keeping the user interface simple and intuitive. Additionally, designing the app to be accessible for both rural and urban users took several iterations and feedback sessions. Overcoming these challenges helped me improve my problem-solving skills, learn AI integration, and understand the importance of user-centered design.

Take-home task

https://github.com/harshad3474-commits/AI-Blood-Donor-Emergency-Help-Finder_project-Report

AI Tools you can use for Step 8:

Canva AI: You can use this to design your pitch document. Download your pitch document as a PDF file and upload on GitHub